

Solved selected problems of Modern Physics for Scientists and Engineers - John Taylor

Franco Zacco

Chapter 11 - Atomic Transitions and Radiation

Solution. 11.6

(a) We know that

$$\frac{dr}{dt} = \frac{dr}{dE} \frac{dE}{dt}$$

Using that $E = -ke^2/2r$ we have that

$$\frac{dE}{dr} = \frac{ke^2}{2r^2} \quad \text{so} \quad \frac{dr}{dE} = \frac{2r^2}{ke^2}$$

Then dr/dt when $r = a_B$ is

$$\frac{dr}{dt} = -\frac{2a_B^2}{ke^2}P = -\frac{2 \cdot 0.0529^2}{1.44} \cdot 2.9 \times 10^{11} = -1.13 \times 10^9 \text{ nm/s}$$

(b) Assuming that dr/dt remains constant, we have that $\Delta t = \Delta r/C$ where $C = -2a_B^2P/ke^2$, hence

$$\Delta t = -\frac{[0 - a_B]}{2a_B^2P/ke^2} = 4.68 \times 10^{-11} \text{ s}$$

The value estimated is in the order of the actual value.

□

Solution. 11.11

- (a) Let a classical point charge q of mass m be in circular orbit of radius r around a fixed charge Q .

Equation (11.1) states that

$$P = \frac{2kq^2a^2}{3c^3}$$

We know the charge's acceleration is given by

$$a = \frac{F}{m} = \frac{kqQ}{mr^2}$$

Hence

$$P = \frac{2kq^2}{3c^3} \frac{k^2q^2Q^2}{m^2r^4} = \frac{2k^3q^4Q^2}{3m^2c^3r^4}$$

- (b) If we double q then P becomes

$$P = \frac{2k^3(2q)^4Q^2}{3m^2c^3r^4} = 16 \cdot \frac{2k^3q^4Q^2}{3m^2c^3r^4}$$

So P is changed by a factor of 16.

- (c) If we double Q then P becomes

$$P = \frac{2k^3q^4(2Q)^2}{3m^2c^3r^4} = 4 \cdot \frac{2k^3q^4Q^2}{3m^2c^3r^4}$$

So P is changed by a factor of 4.

□

Solution. 11.12

- (a) We know that the velocity of an electron orbiting in a Hydrogen atom is

$$v = \sqrt{\frac{ke^2}{mn^2r}}$$

So if $r = a_B$ we get that

$$v_1 = \sqrt{\frac{1.44}{511000 \cdot 0.0529}} = 0.0072c = 2158505.28 \text{ m/s}$$

Then the orbital frequency in the $n = 1$ Bohr orbit of a Hydrogen atom is

$$f_{orb}(1) = \frac{v_1}{2\pi a_B} = 6.49 \times 10^{15} \text{ Hz}$$

If the electron is in the $n = 2$ Bohr orbit then we have that

$$v_2 = \sqrt{\frac{1.44}{511000 \cdot 2^2 \cdot 0.0529}} = 0.0036c = 1094040.96 \text{ m/s}$$

And that

$$f_{orb}(2) = \frac{v_2}{2\pi(2^2 a_B)} = 0.822 \times 10^{15} \text{ Hz}$$

- (b) A electron at $n = 2$ will have an energy of

$$E_2 = -\frac{E_R}{2^2} = -3.4 \text{ eV}$$

Then the transition from $n = 2$ to $n = 1$ emits a photon with an energy of

$$E_2 - E_1 = -3.4 - (-13.6) = 10.2 \text{ eV}$$

And by the De Broglie relation this photon will have a frequency of

$$f_\gamma(2 \rightarrow 1) = \frac{E_2 - E_1}{h} = \frac{10.2}{4.136 \times 10^{-15}} = 2.46 \times 10^{15} \text{ Hz}$$

We see that

$$f_{avg} = \frac{f_{orb}(1) + f_{orb}(2)}{2} = 3.656 \times 10^{15} \text{ Hz}$$

Also

$$f_{diff} = f_{orb}(1) - f_{orb}(2) = 5.668 \times 10^{15} \text{ Hz}$$

We note that $f_\gamma(2 \rightarrow 1)$ is not equal to any of these values.

(c) We see that

$$\begin{aligned}
f_{orb}(n) &= \frac{1}{2\pi n^2 a_B} \sqrt{\frac{ke^2}{mn^2 a_B}} \\
&= \frac{ke^2 m}{2\pi n^2 \hbar^2} \sqrt{\frac{(ke^2)^2}{n^2 \hbar^2}} \\
&= \frac{ke^2 m}{2\pi n^2 \hbar^2} \frac{ke^2}{n \hbar} \\
&= \frac{m(ke^2)^2}{2\pi n^3 \hbar^3}
\end{aligned}$$

Since $E_R = m(ke^2)^2/2\hbar^2$ we get that

$$f_{orb}(n) = \frac{E_R}{\pi n^3 \hbar}$$

On the other hand, we have that

$$\begin{aligned}
f_\gamma(n \rightarrow n-1) &= \frac{1}{h} \left[-\frac{E_R}{n^2} + \frac{E_R}{(n-1)^2} \right] \\
&= \frac{E_R}{h} \frac{2n-1}{n^2(n-1)^2}
\end{aligned}$$

As $n \rightarrow \infty$ then $n-1 \approx n$ and $2n \gg 1$ hence

$$f_\gamma(n \rightarrow n-1) \approx \frac{E_R}{h} \frac{2n}{n^4} = \frac{E_R}{2\pi \hbar} \frac{2}{n^3} = \frac{E_R}{\pi n^3 \hbar}$$

□

Solution. 11.14 Let us estimate the lifetime of a classical He^+ ion starting in the $n = 1$ Bohr orbit of He .

For Hydrogen-like ions we have that

$$E = -\frac{Zke^2}{2r}$$

So

$$\frac{dE}{dr} = \frac{Zke^2}{2r^2} \quad \text{or} \quad \frac{dr}{dE} = \frac{2r^2}{Zke^2}$$

Then in this case we get that

$$\frac{dr}{dE} = \frac{r^2}{ke^2}$$

Where we used that $Z = 2$. Now dr/dt with $r = a_B/Z$ gives us

$$\frac{dr}{dt} = -\frac{a_B^2}{Z^2ke^2}Z^2P = -\frac{0.0529^2}{1.44} \cdot 2.9 \times 10^{11} = -5.63 \times 10^8 \text{ nm/s}$$

Where we used that now P is actually Z^2P because the force involved in the equation for P is now $F = Zke^2/r^2$.

Assuming that dr/dt remains constant we see that

$$\Delta t = \frac{\Delta r}{dr/dt} = -\frac{0 - a_B/Z}{a_B^2P/ke^2} = \frac{0.0529}{2 \cdot 5.63 \times 10^8} = 4.69 \times 10^{-11} \text{ s}$$

Given that $Z = 2$ we get that essentially it is the same lifetime for an electron in a Hydrogen atom. $\square$

Solution. 11.15

Taking equation (11.49) and using that

$$\frac{dr}{dE} = \frac{2r^2}{ke^2}$$

We get that dr/dt is

$$\frac{dr}{dt} = \frac{dr}{dE} \frac{dE}{dt} = -\frac{2r^2}{ke^2} P = -\frac{2r^2}{ke^2} \frac{2(ke^2)^3 c}{3(mc^2)^2 r^4} = -\frac{4(ke^2)^2 c}{3(mc^2)^2 r^2}$$

In this case we do not assume that $r = a_B$.

Now to find the exact Δt , we integrate from a_B to 0 as follow

$$\begin{aligned} \Delta t &= \int_{a_B}^0 \frac{dr}{dr/dt} \\ &= -\frac{3(mc^2)^2}{4(ke^2)^2 c} \int_{a_B}^0 r^2 dr \\ &= -\frac{3(mc^2)^2}{4(ke^2)^2 c} \left[\frac{r^3}{3} \right]_{a_B}^0 \\ &= \frac{(mc^2)^2 a_B^3}{4(ke^2)^2 c} \end{aligned}$$

Therefore

$$\begin{aligned} \Delta t &= \frac{(mc^2)^2 a_B^3}{4(ke^2)^2 c} \\ &= \frac{(5.1 \times 10^5)^2 \cdot 0.0529^3}{4 \cdot (1.44)^2 \cdot 3 \times 10^{17}} \\ &= 1.54 \times 10^{-11} \text{ s} \end{aligned}$$

The value we get is of the same order as the approximation of Problem 11.6. $\square$

Solution. 11.17

Let us consider the wave functions

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

Suppose $m \neq n$ then the integral $I = \int_0^a \psi_m(x)^* \psi_n(x) dx$ becomes

$$\begin{aligned} I &= \frac{2}{a} \int_0^a \sin \frac{n\pi x}{a} \sin \frac{m\pi x}{a} dx \\ &= \frac{1}{a} \int_0^a \left[\cos \left(\frac{n\pi x}{a} - \frac{m\pi x}{a} \right) - \cos \left(\frac{n\pi x}{a} + \frac{m\pi x}{a} \right) \right] dx \\ &= \frac{1}{a} \int_0^a \left[\cos \left((n-m) \frac{\pi x}{a} \right) - \cos \left((n+m) \frac{\pi x}{a} \right) \right] dx \\ &= \frac{1}{a} \left[\frac{a \sin \left((n-m) \frac{\pi x}{a} \right)}{\pi(m-n)} - \frac{a \sin \left((n+m) \frac{\pi x}{a} \right)}{\pi(m+n)} \right]_0^a \\ &= \left[\frac{\sin((n-m)\pi)}{\pi(m-n)} - \frac{\sin((n+m)\pi)}{\pi(m+n)} - 0 \right] \\ &= 0 \end{aligned}$$

Where we used that $\sin \alpha \sin \beta = 1/2[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$ and that $\sin((n-m)\pi) = \sin((n+m)\pi) = 0$ because $n \neq m$ and hence for any integer multiple of π we get that $\sin(n\pi) = 0$.

Therefore, the wave functions for the infinite square well satisfy the orthogonality property. $\square$

Solution. 11.22

- (a) The probability of an electron to be found in the ground state when it was originally in the first excited state is given by

$$\begin{aligned}
 P(2 \rightarrow 1) &= \left(\frac{e\mathcal{E}\Delta t}{\hbar} \right)^2 \left| \frac{2}{a} \int_{-a/2}^{a/2} \cos \frac{\pi x}{a} x \sin \frac{2\pi x}{a} \right|^2 \\
 &= \left(\frac{e\mathcal{E}\Delta t}{\hbar} \right)^2 \left| \frac{2}{a} \int_{-a/2}^{a/2} \sin \frac{2\pi x}{a} x \cos \frac{\pi x}{a} \right|^2 \\
 &= P(1 \rightarrow 2) \\
 &= \left(\frac{e\mathcal{E}\Delta t}{\hbar} \frac{16a}{9\pi^2} \right)^2
 \end{aligned}$$

- (b) The probability $P(2 \rightarrow 3)$ is given by

$$\begin{aligned}
 P(2 \rightarrow 3) &= \left(\frac{e\mathcal{E}\Delta t}{\hbar} \right)^2 \left| \frac{2}{a} \int_{-a/2}^{a/2} \cos \frac{3\pi x}{a} x \sin \frac{2\pi x}{a} \right|^2 \\
 &= \left(\frac{e\mathcal{E}\Delta t}{\hbar} \frac{48a}{25\pi^2} \right)^2
 \end{aligned}$$

- (c) The probability $P(2 \rightarrow 4)$ is equal to 0 because both 2 and 4 are even, and to the extent of the approximation we are assuming the integral vanishes, so it's a "forbidden transition".

The probability $P(2 \rightarrow 5)$ is given by

$$\begin{aligned}
 P(2 \rightarrow 5) &= \left(\frac{e\mathcal{E}\Delta t}{\hbar} \right)^2 \left| \frac{2}{a} \int_{-a/2}^{a/2} \cos \frac{5\pi x}{a} x \sin \frac{2\pi x}{a} \right|^2 \\
 &= \left(\frac{e\mathcal{E}\Delta t}{\hbar} \frac{80a}{441\pi^2} \right)^2
 \end{aligned}$$

So we see that the transition $2 \rightarrow 3$ and the transition $2 \rightarrow 1$ have almost the same probability since $(16/9)^2 = 3.16$ and $(48/25)^2 = 3.68$. The transition $2 \rightarrow 5$ has a lot less probability to happen since $(80/441)^2 = 0.032$.

□

Solution. 11.26

(a) Let us consider a wavefunction $\psi_{nlm}(\mathbf{r}) = R_{nl}(r)\Theta_{lm}(\theta)e^{im\phi}$.

In spherical coordinate if we have an point $\mathbf{r} = (r, \theta, \phi)$ then we note that $-\mathbf{r} = (r, \pi - \theta, \phi + \pi)$.

From Table 8.1 if $l = 0$ then $\Theta_{lm}(\theta) = \sqrt{1/4\pi}$ i.e it's a constant so

$$\psi_{nlm}(-\mathbf{r}) = \sqrt{\frac{1}{4\pi}}R_{nl}(r)e^0 = (-1)^0\sqrt{\frac{1}{4\pi}}R_{nl}(r) = (-1)^l\psi_{nlm}(\mathbf{r})$$

If $l = 1$ and $m = 0$ then

$$\begin{aligned}\psi_{nlm}(-\mathbf{r}) &= \sqrt{\frac{3}{4\pi}}\cos(\pi - \theta)R_{nl}(r)e^0 \\ &= -\sqrt{\frac{3}{4\pi}}\cos(\theta)R_{nl}(r) \\ &= (-1)^1\sqrt{\frac{3}{4\pi}}\cos(\theta)R_{nl}(r) \\ &= (-1)^l\psi_{nlm}(\mathbf{r})\end{aligned}$$

If $l = 1$ and $m = 1$ then

$$\begin{aligned}\psi_{nlm}(-\mathbf{r}) &= -\sqrt{\frac{3}{8\pi}}\sin(\pi - \theta)R_{nl}(r)e^{i\phi}e^{i\pi} \\ &= \sqrt{\frac{3}{8\pi}}\sin(\theta)R_{nl}(r)e^{i\phi} \\ &= (-1)^1\left(-\sqrt{\frac{3}{8\pi}}\sin(\theta)R_{nl}(r)e^{i\phi}\right) \\ &= (-1)^l\psi_{nlm}(\mathbf{r})\end{aligned}$$

If $l = 2$ and $m = 0$ then

$$\begin{aligned}\psi_{nlm}(-\mathbf{r}) &= \sqrt{\frac{5}{16\pi}}(3\cos^2(\pi - \theta) - 1)R_{nl}(r)e^0 \\ &= \sqrt{\frac{5}{16\pi}}(3\cos^2(\theta) - 1)R_{nl}(r) \\ &= (-1)^2\sqrt{\frac{5}{16\pi}}(3\cos^2(\theta) - 1)R_{nl}(r) \\ &= (-1)^l\psi_{nlm}(\mathbf{r})\end{aligned}$$

If $l = 2$ and $m = 1$ then

$$\begin{aligned}
\psi_{nlm}(-\mathbf{r}) &= -\sqrt{\frac{15}{8\pi}} \sin(\pi - \theta) \cos(\pi - \theta) R_{nl}(r) e^{i\phi} e^{i\pi} \\
&= -\sqrt{\frac{15}{8\pi}} \sin(\theta) \cos(\theta) R_{nl}(r) e^{i\phi} \\
&= (-1)^2 - \sqrt{\frac{15}{8\pi}} \sin(\theta) \cos(\theta) R_{nl}(r) e^{i\phi} \\
&= (-1)^l \psi_{nlm}(\mathbf{r})
\end{aligned}$$

If $l = 2$ and $m = 2$ then

$$\begin{aligned}
\psi_{nlm}(-\mathbf{r}) &= \sqrt{\frac{15}{32\pi}} \sin^2(\pi - \theta) R_{nl}(r) e^{i2\phi} e^{i2\pi} \\
&= \sqrt{\frac{15}{32\pi}} \sin^2(\theta) R_{nl}(r) e^{i2\phi} \\
&= (-1)^2 \sqrt{\frac{15}{32\pi}} \sin^2(\theta) R_{nl}(r) e^{i2\phi} \\
&= (-1)^l \psi_{nlm}(\mathbf{r})
\end{aligned}$$

Therefore the equality $\psi_{nlm}(-\mathbf{r}) = (-1)^l \psi_{nlm}(\mathbf{r})$ holds for every function of Table 8.1.

- (b) Let us consider the probability of a radiative transition $n, l, m \rightarrow n', l', m'$

$$P(n, l, m \rightarrow n', l', m') \propto \mathcal{E}_0^2 \left| \int \psi_{n'l'm'}^* x \psi_{nlm} dV \right|^2$$

Let us change variables $\mathbf{r} = -\mathbf{r}$ and hence $x = -x$ then we see that the integral becomes

$$\begin{aligned}
\int \psi_{n'l'm'}^*(\mathbf{r}) x \psi_{nlm}(\mathbf{r}) dV &= \int \psi_{n'l'm'}^*(-\mathbf{r}) (-x) \psi_{nlm}(-\mathbf{r}) dV \\
&= \int (-1)^{l'} \psi_{n'l'm'}^*(\mathbf{r}) (-x) (-1)^l \psi_{nlm}(\mathbf{r}) dV
\end{aligned}$$

If we consider that $l = l'$ then

$$\begin{aligned}
\int \psi_{n'l'm'}^*(\mathbf{r}) x \psi_{nlm}(\mathbf{r}) dV &= \int (-1)^{2l+1} \psi_{n'l'm'}^*(\mathbf{r}) x \psi_{nlm}(\mathbf{r}) dV \\
&= - \int \psi_{n'l'm'}^*(\mathbf{r}) x \psi_{nlm}(\mathbf{r}) dV
\end{aligned}$$

We used that $2l + 1$ is an odd number so $(-1)^{2l+1} = -1$. Therefore must be that

$$\int \psi_{n'l'm'}^*(\mathbf{r}) x \psi_{nlm}(\mathbf{r}) dV = 0$$

And hence $P(n, l, m \rightarrow n', l, m') = 0$.

□

Solution. 11.27

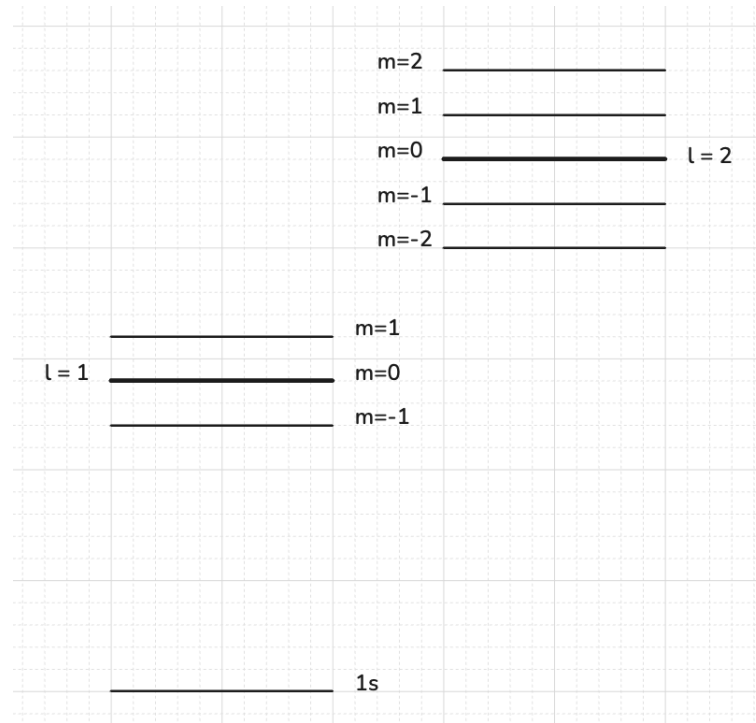
There is an metastable state of ^{180}Ta which is called $^{180m1}\text{Ta}$ where the nucleus is in an excited state. Normally the metastable states have a short lifetime because of the unstability of the nucleus and as a result they show β decay.

In the case of $^{180m1}\text{Ta}$ the total angular momentum number j_{tot} is bigger than 1 so the possible transitions make Δj_{tot} bigger than 1 and hence the transitions and the β decay are improbable.

Therefore $^{180m1}\text{Ta}$ is an exceptionally stable state of ^{180}Ta . $\square$

Solution. 11.28

(a) Below we show a sketch of the resulting levels.



(b) Let us consider the initial level to be m_i and the final level m_f , then we can count the different transitions as follows.

If $m_i = 2$ then the possible transitions are

$$2 - 1 = 1$$

$$2 - 0 = 2$$

$$2 - (-1) = 3$$

If $m_i = 1$ then the possible transitions are

$$1 - 1 = 0$$

$$1 - 0 = 1 \text{ (repeated)}$$

$$1 - (-1) = 2 \text{ (repeated)}$$

If $m_i = 0$ then the possible transitions are

$$0 - 1 = -1$$

$$0 - 0 = 0 \text{ (repeated)}$$

$$0 - (-1) = 1 \text{ (repeated)}$$

If $m_i = -1$ then the possible transitions are

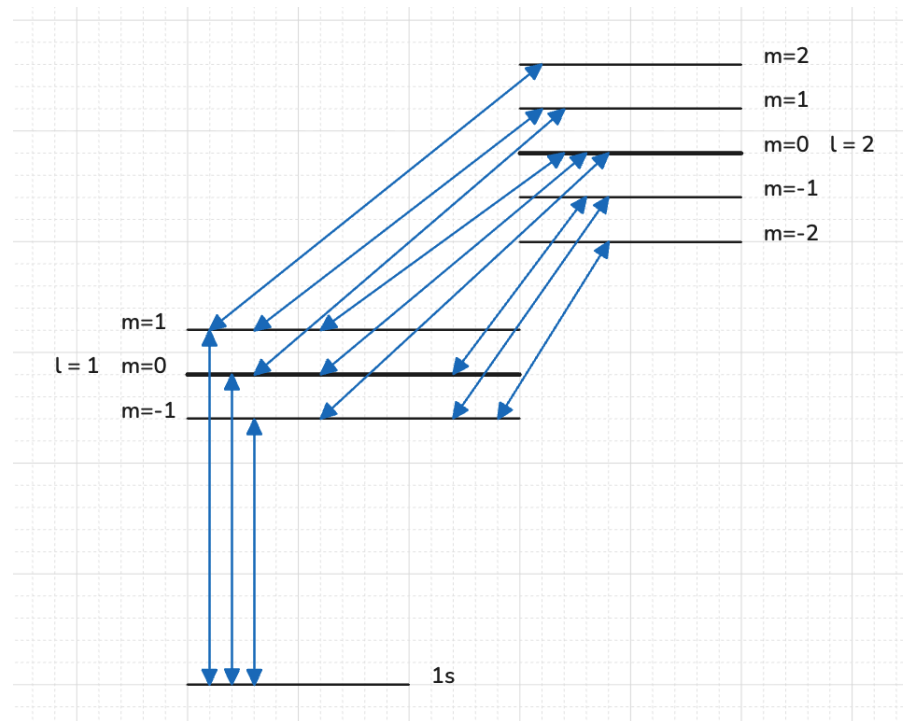
$$\begin{aligned}(-1) - 1 &= -2 \\(-1) - 0 &= -1 \text{ (repeated)} \\(-1) - (-1) &= 0 \text{ (repeated)}\end{aligned}$$

If $m_i = -2$ then the possible transitions are

$$\begin{aligned}(-2) - 1 &= -3 \\(-2) - 0 &= -2 \text{ (repeated)} \\(-2) - (-1) &= -1 \text{ (repeated)}\end{aligned}$$

Therefore there are 7 distinct transitions.

(c) The allowed transitions that satisfy the selection rules are



(d) The photon frequencies that will result from the transitions from the level $l = 1$ to the level $l = 2$ (or viceversa) depend only on the difference $m_i - m_f$ and since Δm can only be 0 or ± 1 then there are only 3 allowed photon frequencies when the electron transition between the levels $l = 1$ and $l = 2$.

Additionally, there are 3 more distinct possible photon frequencies when the electron transition from the level $l = 1$ to the ground level. Therefore there are 6 distinct photon frequencies.

□

Solution. 11.30

Equation (8.98) states that the wavefunction is of the form

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r)\Theta_{lm}(\theta)e^{im\phi}$$

Let us write the probability integral in spherical coordinates for the x component of the radiation

$$\int_0^\infty \int_0^\pi \int_0^{2\pi} (R_{n'l'}(r)\Theta_{l'm'}(\theta)e^{-im'\phi})r \sin \theta \cos \phi (R_{nl}(r)\Theta_{lm}(\theta)e^{im\phi}) r^2 \sin \theta dr d\theta d\phi$$

For the ϕ variable the integral is

$$\int_0^{2\pi} \cos \phi e^{-im'\phi} e^{im\phi} d\phi = \int_0^{2\pi} \cos \phi e^{i\phi(m-m')} d\phi$$

Naming $m - m' = n$ and assuming that $|n| \geq 2$ we get that

$$\int_0^{2\pi} \cos \phi e^{in\phi} d\phi = \frac{in(1 - e^{2i\pi n})}{n^2 - 1}$$

We note that $e^{2i\pi n} = 1$ for $|n| = 2, 3, \dots$ so the integral is 0.

In the case $n = \pm 1$ i.e. $\Delta m = \pm 1$ we have that

$$\int_0^{2\pi} \cos \phi e^{\pm i\phi} d\phi = \pi$$

If $\Delta m = 0$ the integral becomes

$$\int_0^{2\pi} \cos \phi d\phi = 0$$

For the y component of the radiation, the ϕ integral is

$$\int_0^{2\pi} \sin \phi e^{i\phi(m-m')} d\phi$$

Again, this integral is 0 for any $|\Delta m| = 0, 2, 3, \dots$ and it's equal to $i\pi$ for $\Delta m = \pm 1$. Finally, the ϕ -integral for the z component of the radiation is

$$\int_0^{2\pi} e^{i\phi(m-m')} d\phi$$

Which is 0 for $|\Delta m| = 1, 2, 3, \dots$ but it's equal to 2π when $\Delta m = 0$.

Therefore the average of the three integrals is non-zero only when $\Delta m = 0$ or ± 1 . $\square$

Solution. 11.31

- (a) We know that to interact with the atom, the orbital angular momentum of the photon must be of order $a_B p$ or less, where p is the linear momentum of the photon, i.e.

$$L_\gamma \leq a_B p$$

Then

$$L_\gamma \leq a_B \frac{E_R}{c} = a_B \frac{ke^2}{2a_B c} = \frac{ke^2 \hbar}{2c \hbar} = \frac{\alpha}{2} \hbar \leq \alpha \hbar$$

Where we used that $p = E_R/c$, $E_R = ke^2/2a_B$ and that $\alpha = ke^2/c\hbar$.

- (b) Taking the z component of $\mathbf{L}' = \mathbf{L} + \mathbf{S}_\gamma$ we see that

$$L'_z = L_z + S_{\gamma z}$$

We know that the maximum value for L_z is $l\hbar$. The allowed values for $S_{\gamma z}$ are $\hbar, 0, -\hbar$ so $L'_z = l'\hbar$ will be maximum when $L_z = l\hbar$ and $S_{\gamma z} = \hbar$ so

$$\begin{aligned} l'\hbar &= l\hbar + \hbar \\ l' &= l + 1 \end{aligned}$$

On the other hand, L'_z will be minimum when $L_z = l\hbar$ and $S_{\gamma z} = -\hbar$, hence

$$l' = l - 1$$

Given that the selection rule says that $l' = l$ cannot happen then $S_{\gamma z} = 0$ is forbidden. Therefore must be that $\Delta l = l' - l = \pm 1$.

□

Solution. 11.33

- (a) We know that the energy of a single photon is $E = hc/\lambda$, since the laser has $\lambda = 532 \text{ nm}$ then

$$E = \frac{6.63 \times 10^{-34} \text{ Js} \cdot 2.998 \times 10^8 \text{ m/s}}{532 \times 10^{-9} \text{ m}} = 3.736 \times 10^{-19} \text{ J}$$

Each pulse has 0.3 J then the number of photons sent is

$$\frac{0.3}{3.736 \times 10^{-19}} = 8.03 \times 10^{17} \text{ photons}$$

- (b) The laser beam has a 5 km diameter on the moon so the area of the beam is $A_l = \pi 5^2/4 = 19.63 \text{ km}^2$.

The area of each mirror is $\pi(4 \times 10^{-5})^2/4 = 1.25 \times 10^{-9} \text{ km}^2$ so the total area of the 300 mirrors is $A_m = 300 \cdot 1.25 \times 10^{-9} = 3.75 \times 10^{-7} \text{ km}^2$.

Then the ratio of photons coming back to Earth is

$$\frac{3.75 \times 10^{-7}}{19.63} = 1.91 \times 10^{-8}$$

So the number of photons reaching the Earth would be

$$1.91 \times 10^{-8} \cdot 8.03 \times 10^{17} \text{ photons} = 15.33 \times 10^9 \text{ photons}$$

- (c) The angular divergence is given by

$$\delta\theta \approx \frac{\lambda}{d} = \frac{532 \times 10^{-9}}{0.04 \cdot 300} = 4.43 \times 10^{-8}$$

- (d) The diameter of the reflected beam is

$$d' = 2R = 2L\delta\theta = 2 \cdot 4 \times 10^8 \cdot 4.43 \times 10^{-8} = 35.44 \text{ m}$$

- (e) The area of the beam coming to Earth is $\pi d'^2/4 = 986.45 \text{ m}^2$ and the area of the telescope is $\pi 1^2/4 = 0.78 \text{ m}^2$ so the fraction of light captured by the telescope is

$$\frac{0.78}{986.45} = 0.0008 = 0.08 \%$$

- (f) Given that only 0.08% of the light is captured by the telescope then the number of photons captured is

$$15.33 \times 10^9 \cdot 0.0008 = 12.264 \times 10^6 \text{ photons}$$

□

Solution. 11.34

Let us consider an infinite square well of width $a = 1$, centered on the origin and let

$$\psi_n(x) = \begin{cases} \sqrt{2} \cos n\pi x & (n \text{ is odd}) \\ \sqrt{2} \sin n\pi x & (n \text{ is even}) \end{cases}$$

be the stationary state wave functions. Let also

$$\psi(x) = \begin{cases} 1 & x \in [-1/2, 1/2] \\ 0 & x \in (-\infty, -1/2) \cup (1/2, \infty) \end{cases}$$

(a) Let us compute the coefficients A_n as follows.

If n is odd we have that

$$\begin{aligned} A_n &= \int_{-\infty}^{\infty} \psi_n(x)^* \psi(x) dx \\ &= \sqrt{2} \int_{-1/2}^{1/2} \cos n\pi x dx \\ &= \frac{\sqrt{2}}{\pi n} \left[\sin n\pi x \right]_{-1/2}^{1/2} \\ &= \frac{\sqrt{2}}{\pi n} \left[\sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right] \\ &= \frac{2\sqrt{2}}{\pi n} \sin \frac{n\pi}{2} \end{aligned}$$

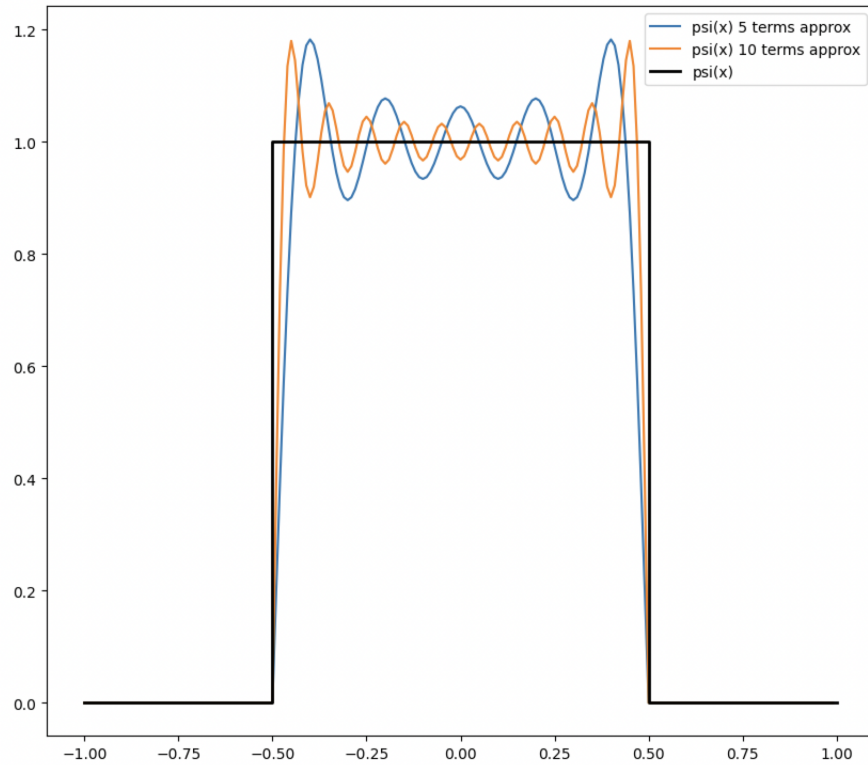
If n is even then we have that

$$\begin{aligned} A_n &= \int_{-\infty}^{\infty} \psi_n(x)^* \psi(x) dx \\ &= \sqrt{2} \int_{-1/2}^{1/2} \sin n\pi x dx \\ &= -\frac{\sqrt{2}}{\pi n} \left[\cos n\pi x \right]_{-1/2}^{1/2} \\ &= -\frac{\sqrt{2}}{\pi n} \left[\cos \frac{n\pi}{2} - \cos \frac{n\pi}{2} \right] \\ &= 0 \end{aligned}$$

(b)-(c) The first 5 nonzero terms of the series are

$$\begin{aligned} A_1\psi_1(x) &= \frac{4}{\pi} \cos \pi x & A_3\psi_3(x) &= -\frac{4}{3\pi} \cos 3\pi x \\ A_5\psi_5(x) &= \frac{4}{5\pi} \cos 5\pi x & A_7\psi_7(x) &= -\frac{4}{7\pi} \cos 7\pi x \\ A_9\psi_9(x) &= \frac{4}{9\pi} \cos 9\pi x \end{aligned}$$

Plotting the sum of the 5 nonzero terms and the 10 nonzero terms we get that



□

Solution. 11.35

Let us consider now

$$\psi(x) = \sqrt{2}|(1 - 2x)|$$

(a) Let us compute the coefficients A_n as follows.

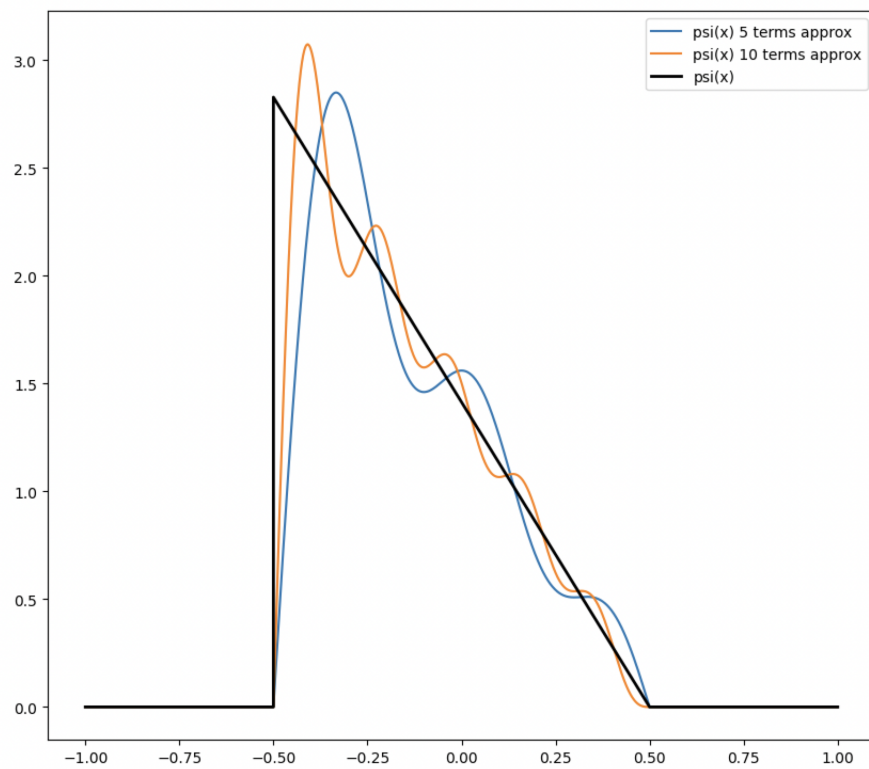
If n is odd we have that

$$\begin{aligned} A_n &= \int_{-\infty}^{\infty} \psi_n(x)^* \psi(x) \, dx \\ &= 2 \int_{-1/2}^{1/2} (1 - 2x) \cos n\pi x \, dx \\ &= \frac{4}{\pi n} \sin \frac{n\pi}{2} \end{aligned}$$

If n is even then we have that

$$\begin{aligned} A_n &= \int_{-\infty}^{\infty} \psi_n(x)^* \psi(x) \, dx \\ &= 2 \int_{-1/2}^{1/2} (1 - 2x) \sin n\pi x \, dx \\ &= \frac{4}{\pi^2 n^2} \left[\pi n \cos \frac{n\pi}{2} - 2 \sin \frac{n\pi}{2} \right] \end{aligned}$$

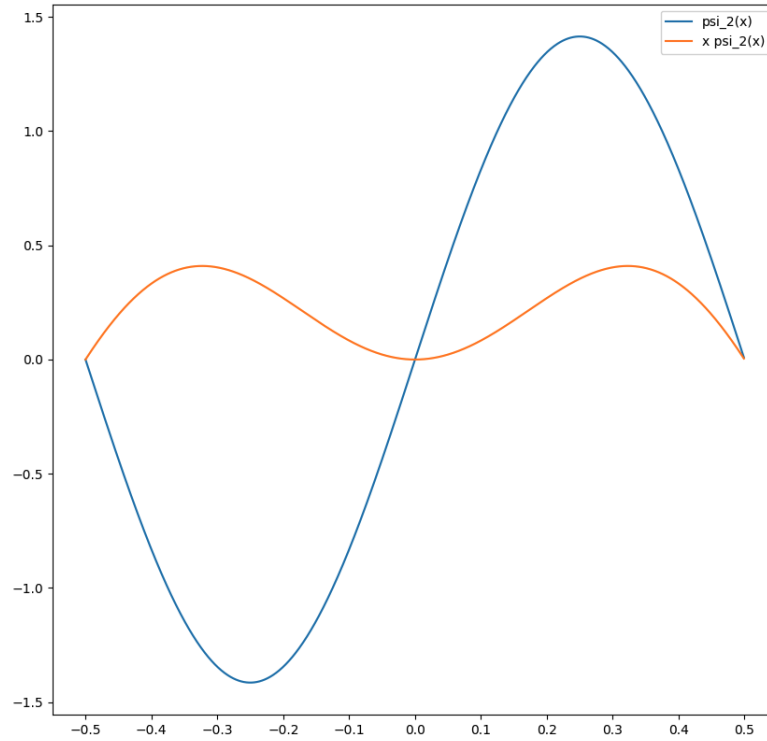
(b)-(c) Plotting the sum of the 5 nonzero terms and the 10 nonzero terms we get that



□

Solution. 11.36

(a) Below we plot $\psi_2(x)$ and $x\psi_2(x)$ assuming $a = 1$.



(b) Let us expand the function $x\psi_2(x)$, then the coefficients A_n if n is odd are given by

$$\begin{aligned}
 A_n &= \int_{-\infty}^{\infty} \psi_n(x)^* x\psi_2(x) dx \\
 &= \frac{2}{a} \int_{-a/2}^{a/2} x \cos \frac{n\pi x}{a} \sin \frac{2\pi x}{a} dx \\
 &= -\frac{4a[\pi(n^2 - 4) \cos(n\pi/2) - 4n \sin(n\pi/2)]}{\pi^2(n-2)^2(n+2)^2}
 \end{aligned}$$

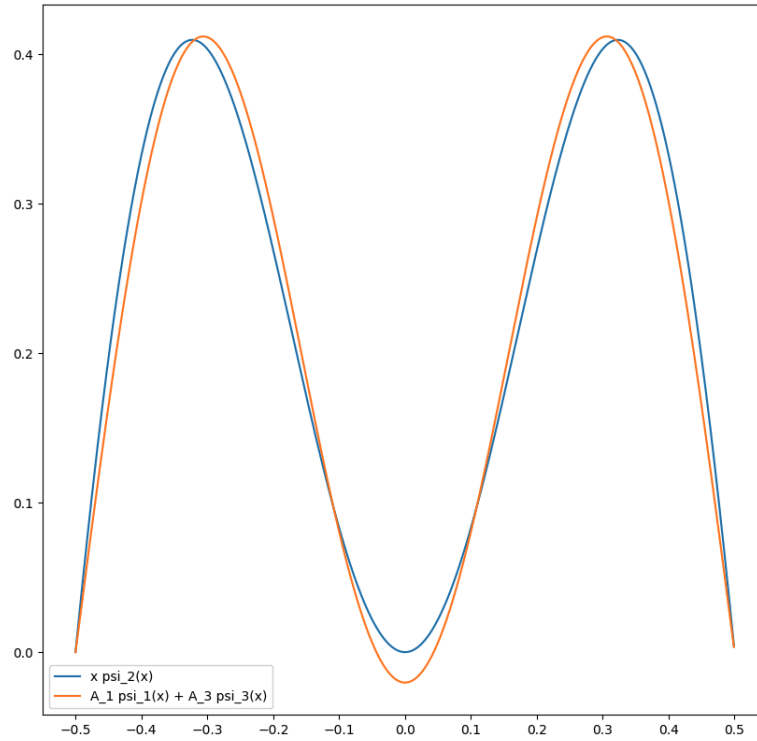
If n is even

$$A_n = \frac{2}{a} \int_{-a/2}^{a/2} x \sin \frac{n\pi x}{a} \sin \frac{2\pi x}{a} dx = 0$$

Therefore to a good extent we can approximate $x\psi_2(x)$ as

$$\begin{aligned}
 x\psi_2(x) &\approx A_1\psi_1(x) + A_3\psi_3(x) \\
 &= \frac{4a[3\pi \cos(\pi/2) + 4\sin(\pi/2)]}{9\pi^2}\psi_1(x) \\
 &\quad - \frac{4a[5\pi \cos(3\pi/2) - 12\sin(3\pi/2)]}{25\pi^2}\psi_3(x) \\
 &= \frac{16a}{9\pi^2}\psi_1(x) - \frac{48a}{25\pi^2}\psi_3(x) \\
 &= \sqrt{2/a} \left[\frac{16a}{9\pi^2} \cos \frac{\pi x}{a} - \frac{48a}{25\pi^2} \cos \frac{3\pi x}{a} \right]
 \end{aligned}$$

If we assume $a = 1$ we can plot the approximation against the real value as follows



The physical significance of this is that adding a perturbation to the state ψ_2 adds a term to the wave function that closely approximates $A\psi_1 + B\psi_3$.

□